Geometric Target Repair: Proven Bounds and the Remaining Depth Problem

Balance Trees project

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Abstract

We analyze the geometric-target local-repair implementation in this repository. Each successful insertion preserves binary-search-tree order and takes at most two rotations. The implementation draws fewer than two fair bits in expectation and stores no per-node balance metadata. Its expected ancestor-walk cost is controlled precisely by the insertion depth's excess over $\lceil \log_2 n \rceil$. Neither a constant bound on that excess nor logarithmic expected depth holds against adaptive insertion sequences, although both hold for monotone sequences. We give a distribution-preserving sampler that is optimal in expected fair bits, and a general logarithmic lower bound for some node's expected depth.

1 Model and algorithm

Keys are distinct and totally ordered; a duplicate is rejected before policy state or random bits change. Start with an empty binary search tree (BST). Depth counts edges from the root. A successful insertion first attaches the new key as a leaf v, at depth d, and sets n to the number of keys including v. Write

$$k = \lceil \log_2 n \rceil, \qquad L = k + 1.$$

The strategy has a global successful-insertion counter; each node contains only its key and structural child and parent links. The count needs $\Theta(\log n)$ bits (one word in the usual word-RAM model). We count comparisons for the BST search separately. A rotation and a parent-pointer step have constant cost; a fair-bit request costs one. Exact k is obtained in constant word operations, as with the implementation's integer bit length. Random-generator state is external to tree nodes and is not free storage in an implementation.

Draw independent fair bits until the first zero or L bits have been read. If m ones were read before stopping, set the target $T = \max(0, k-m)$. If $d \le T$, do nothing; if $d = T+1$, rotate v up once. Otherwise walk from v to its ancestor x at depth $T+2$. Let p and g be the parent and grandparent of x. If g, p, x form a straight path, rotate p up once. If the path bends, rotate x up twice. All rotations are ordinary BST rotations. This is the repository's `geometric_target_local_repair` and `target_repair` code; it is distinct from the geometric stopping rules studied by Brodal [1].

Each bit in the mathematical model is independent of the preceding history and of the current key, conditional on that history. We permit the next key to be chosen adaptively from the observed past, before the new bits are drawn. This covers fixed sequences. Python's seeded pseudorandom generator is deterministic; the probability statements below describe the ideal fair-bit model, not a cryptographic or information-theoretic guarantee about its finite seed.

2 Deterministic properties

Theorem 1. *After every successful insertion the result is a BST containing exactly the previous keys and the new key, and at most two rotations have occurred. No per-node balancing information is written. The claim holds for every bit stream and for both fixed and adaptive insertion sequences.*

Proof. Ordinary leaf insertion adds the new key in its unique search position. A single rotation preserves the inorder sequence: for example, rotating a left child p above its parent g replaces the local order A, p, B, g, C by the same order; the other orientation is symmetric. Rotations neither add nor remove keys and preserve parent/child connectivity. The depth cases use zero, one, one, or two rotations respectively. In the last case x has a parent and grandparent because $T \geq 0$ and x is at depth $T + 2$; in the bent case the first promotion leaves x directly below g, so the second is defined. The only persistent policy state is the global count, and rotations update structural links only. □

Lemma 2 (Exact visit count)**.** *Let V be the number of* counted ancestor visits *in the implementation's repair routine, including the final inspected ancestor. Then, pointwise,*

$$V = (d - T - 1)_+ .$$

Proof. If $d \leq T + 1$ the routine records no visits, and the right-hand side is zero. Otherwise it traverses $d - T - 2$ parent links and counts each departure, then counts the selected ancestor once. Thus it records $d - T - 1$ visits. Other constant work, including the single-rotation case, is not counted as an ancestor visit. □

3 Exact random-bit and repair costs

Theorem 3 (Target distribution and bits)**.** *For $k \geq 0$, the target and number B of requested bits satisfy*

$$\mathbb{P}(T = 0) = 2^{-k}, \qquad \mathbb{P}(T = t) = 2^{-(k-t+1)} \quad (1 \leq t \leq k),$$
$$\mathbb{E}B = 2 - 2^{-k} < 2, \qquad \mathbb{E}(k - T) = 1 - 2^{-k}.$$

The statements hold conditional on any prior history and current key.

Proof. For $1 \leq t \leq k$, the first $k-t$ bits must be one and the next zero. Target zero occurs precisely when the first k bits are all one, irrespective of the following bit; for $k = 0$ this event is certain. The ith bit is requested exactly when the preceding $i-1$ bits are all one, for $1 \leq i \leq k+1$. Summing these probabilities gives $\mathbb{E}B = \sum_{j=0}^{k} 2^{-j} = 2 - 2^{-k}$. Finally $k - T = \min(G, k)$, where G is the number of leading ones in an unbounded stream, so the tail-sum formula gives $\mathbb{E}(k-T) = \sum_{j=1}^{k} 2^{-j} = 1 - 2^{-k}$. Conditional independence makes the same calculation valid after every history. □

Theorem 4 (Exact and bounded expected walking)**.** *Let D be the new leaf's depth before repair. Conditional on a history and current key, $D = d$ is fixed and the newly sampled T is independent of it. If $d \geq k + 1$, then*

$$\mathbb{E}[V \mid D = d] = d - k - 2^{-k}.$$

For random D, including an adaptively chosen next key,

$$\mathbb{E}\,(D - k - 1)_+ \;\leq\; \mathbb{E}V \;\leq\; \mathbb{E}\,(D - k - 1)_+ + 1 - 2^{-k}.$$

Consequently a uniform $O(1)$ expected repair-time bound is equivalent to a uniform $O(1)$ bound on $\mathbb{E}\,(D - k - 1)_+$, under the stated cost model.

Proof. For $d \geq k+1$, $d-T-1 \geq 0$ for every target, so Lemma 2 and Theorem 3 give $\mathbb{E}V = d-k-1+\mathbb{E}(k-T) = d-k-2^{-k}$. For any d and $T \leq k$,

$$(d-k-1)_+ \leq (d-T-1)_+ \leq (d-k-1)_+ + (k-T).$$

Take conditional expectations and then average over the prior history. Sampling and constant rotation work have expected $O(1)$ cost, while the parent loop costs $\Theta(V)$ up to additive constants. The two inequalities therefore prove the equivalence. An $O(\log n)$ bound on $\mathbb{E}D$ alone does not imply the required additive-excess bound. □

Theorem 5 (A reachable conditional-time counterexample)**.** *There is no uniform constant bound on expected repair time conditional on every reachable history, even for a fixed insertion sequence.*

Proof. Insert $1, 2, \ldots, m$ and choose target zero at each insertion. Each such choice has positive probability by Theorem 3, so this finite history has positive probability. Inductively the current maximum is the root: the next key attaches as its right child and rotates above it. The resulting tree is the descending left chain $m, m-1, \ldots, 1$. Now insert zero. Its depth is m, and every target is at most $k = \lceil \log_2(m+1) \rceil$. Therefore Lemma 2 gives $V \geq (m-k-1)_+$ for every fresh target, proving the claim. The fixed sequence is $(1, 2, \ldots, m, 0)$. This rare conditional history does not prove that the *unconditional* expected cost of that sequence, or any other fixed sequence, is superconstant. □

Theorem 6 (Monotone insertion sequences)**.** *For strictly increasing insertions, the maximum key has depth at most $\lceil \log_2 n \rceil$ after every repair, for every sequence of bits. Each insertion uses at most one rotation, and its expected number of counted repair visits, conditional on the past, is at most $1 - 2^{-k} < 1$. The symmetric claims hold for strictly decreasing insertions and the minimum key. Duplicate attempts do not affect these bounds.*

Proof. The root insertion has depth zero and needs no rotation. For $n \geq 2$, assume the previous maximum has depth at most $\lceil \log_2(n-1) \rceil \leq k$. The new maximum attaches as its right child, so its insertion depth d is at most $k+1$. Its entire search path takes right edges. If $d \leq T$, there is no rotation and $d \leq k$. Otherwise the single-leaf case rotates once, and the two-edge case is straight, so it also rotates once. In either case, the rotated child lies on the new maximum's path, so that maximum moves up exactly one level; its final depth is at most k. This proves the induction and rotation bound for every choice of target. Moreover $d \leq k+1$ implies $V = (d-T-1)_+ \leq k-T$, whose conditional expectation is $1-2^{-k}$ by Theorem 3. Mirror left and right for decreasing insertions. This result bounds the current extremal node. □

Theorem 7 (Every node under monotone insertions)**.** *For strictly increasing insertions, fix a key x inserted at time m. After $n \geq m$ insertions its expected depth satisfies*

$$\mathbb{E}[\mathrm{depth}_n(x)] \leq 2\lceil \log_2 m \rceil + \sum_{j=m+1}^{n} \frac{1}{j} = O(\log(n+1)).$$

The same result holds for strictly decreasing insertions by symmetry.

Proof. The *right spine* is the path obtained by following right children from the root. Before each insertion let a be the depth of x. If x is on the right spine, set $b = a$; otherwise let b be the depth of its deepest right-spine ancestor, whose left subtree contains x. By Theorem 6, the old maximum is at depth at most k for the next insertion's k, hence $0 \leq b \leq k$.

The new maximum is attached at depth d on the right spine, with $d > b$. Whenever the sampled target $t < d$, repair rotates the right-spine child at depth $t+1$ above its parent at depth t. For the potential $a+b$ there are three cases. If $t = b$, x moves down one level and b is unchanged. If $t < b$, the branch point moves up one level, so b decreases by one, and the depth a does not increase. In the boundary case where the branch point is the rotated child, its left subtree may move across the rotation and retain its depth; this still gives the stated inequality. If $t > b$, the rotation is strictly below the relevant branch point and neither a nor b changes. If no rotation occurs then $t \geq d > b$, so this last case still applies.

For $1 \leq b \leq k$, Theorem 3 and a geometric sum yield

$$\mathbb{P}(T = b) = \mathbb{P}(T < b) = 2^{-(k-b+1)}.$$

Consequently the conditional expected change of $a+b$ is at most zero. For $b = 0$ it is at most $\mathbb{P}(T = 0) = 2^{-k} \leq 1/j$ at insertion j. Immediately after x is inserted, it is the maximum and lies on the spine; Theorem 6 gives $a = b \leq \lceil \log_2 m \rceil$. Summing the conditional drift bounds over $j = m+1, \ldots, n$, and using $a \leq a+b$, proves the result. The harmonic sum is at most $1 + \ln n$, establishing the asymptotic bound. Mirror left and right for decreasing insertions. □

The following adaptive construction uses the strong law of large numbers and Fatou's lemma; see Williams [4] for both results.

4 An adaptive counterexample

The following lower bound is for the algorithm specified above and the ideal independent fair-bit model. It does not concern the distinct algorithms analyzed by Brodal. The adversary sees the current tree before choosing each next key, and all choices are made before the next insertion's random bits.

Theorem 8 (Linear adaptive depth and repair)**.** *There is an adaptive sequence of distinct rational keys and a key x in the initial prefix such that, after n successful insertions, $\mathbb{E}[\mathrm{depth}_n(x)] = \Omega(n)$. Along an infinite subsequence of insertions, the expected number of counted repair visits is also $\Omega(n)$. The expectations include all random bits, including those in the initial prefix.*

Proof. Throughout, x will be the largest inserted key, so its root-to-node path consists entirely of right edges. Let n include the next key, and write $k = \lceil \log_2 n \rceil$. Assume for the moment that $k \geq 2$ and that the present depth d of x satisfies $d > k+2$. We specify two types of insertion. All required gaps exist because the rational order is dense. For definiteness, choose the midpoint of each bounded gap and one less than the upper endpoint for an unbounded-below gap. The attack depends only on the current tree, including its keys; the fixed schedule below is determined by the number of nodes, which can be counted from that tree.

For an *attack*, let g be the node at depth k on the path to x, and insert a fresh key in g's left subtree, smaller than every key already in that subtree. If the subtree is empty, insert at its root gap. The insertion path is k right edges followed by one or more left edges. There are three cases. With probability $1/2$, $T = k$: a left child of g is promoted once, so g, its right subtree, and x move down one level. This includes the case where the new leaf is the left child of g. With probability $1/4$, $T = k-1$: the selected two-edge path is right–left, so the double rotation leaves g at its old depth and leaves d unchanged. With probability $1/4$, $T \leq k-2$: the selected path is right–right and the single rotation moves x up one level. Thus an attack changes d by $+1, 0, -1$ with respective probabilities $1/2, 1/4, 1/4$.

For a *probe*, insert a fresh key in the gap immediately below x, choosing it within x's left subtree. The insertion path reaches x, and its insertion depth is at least $d+1$. Since $d > k+2$, every target $T \le k$ selects a right–right path strictly above x. Exactly one rotation occurs, moving x up one level. By the exact visit formula, the probe has

$$V = (d_{\text{ins}} - T - 1)_+ \ge d - k.$$

Neither kind of key exceeds x, which remains the maximum.

Fix m in advance. To initialize, insert m distinct keys in decreasing order, starting with 0 and then choosing one less than the current minimum. Afterward, the current maximum is x. At each subsequent step, count the nodes to determine the next size and its position in a fixed schedule of nine attacks followed by one probe. If the current depth of x is at most $k+2$, insert one less than the current minimum; otherwise use the scheduled attack or probe. This is a deterministic function of the current tree and fixed m: it does not remember whether the condition failed previously, inspect priorities or random bits, or know future outcomes. On the event that each sampled target in this prefix is zero, every new minimum rotates to the root, leaving a right chain with $d_m(x) = m - 1$. This event has strictly positive probability for every fixed m.

It remains to choose a fixed m so that the subsequent condition holds forever with positive probability. Couple each future attack to an independent random variable taking values $+1, 0, -1$ with probabilities $1/2, 1/4, 1/4$; assign -1 to each future probe. The coupling, used only in the analysis and not available to the adversary, can read the first two latent fair bits for an attack: zero gives $+1$, one then zero gives 0, and two ones give -1. Latent bits are independent even when a sampler stops before reading them. Let Z_s be the sum of the first s such scheduled increments, with $Z_0 = 0$. The sums of successive ten-increment blocks are independent and have mean $9/4 - 1 = 5/4$. The strong law of large numbers, and the bounded remainder within each block, imply

$$\frac{Z_s}{s} \longrightarrow \frac{1}{8} \quad \text{almost surely.}$$

Consequently

$$M = \sup_{s \ge 1} \big(\lceil \log_2(s+1) \rceil - Z_{s-1} \big)$$

is finite almost surely. Choose a finite integer B with $\mathbb{P}(M \le B) > 0$, then choose a fixed $m \ge 4$ large enough that $m - 1 - \lceil \log_2 m \rceil \ge B + 3$. The initial-prefix event and $M \le B$ are independent and their intersection E has positive probability.

For step s after the prefix, ceiling subadditivity and $m + s \le m(s+1)$ give

$$\lceil \log_2(m+s) \rceil \le \lceil \log_2 m \rceil + \lceil \log_2(s+1) \rceil.$$

Inductively on E, before that step the depth is $m - 1 + Z_{s-1}$, so its excess over the current k is at least 3. Hence the stated attack and probe transitions apply at every step, and on E, $d_{m+s}(x) = m - 1 + Z_s$ and $d_{m+s}(x)/(m+s) \to 1/8$ almost surely. Fatou's lemma yields

$$\liminf_{n \to \infty} \frac{\mathbb{E}[d_n(x)]}{n} \ge \frac{\mathbb{P}(E)}{8} > 0.$$

On the probe subsequence $n = m + 10j$, the bound $V \ge d - k$, applied to the depth just before the probe, and the same argument give

$$\liminf_{j \to \infty} \frac{\mathbb{E}[V_{m+10j}]}{m + 10j} \ge \frac{\mathbb{P}(E)}{8} > 0.$$

Thus both stated lower bounds are unconditional. □

Corollary 9 (Tight adaptive worst case)**.** *Under the model of Theorem 8, the largest expected node depth attainable by an adaptive adversary is* $\Theta(n)$*. The expected repair visits in that theorem are* $\Theta(n)$ *along its probe subsequence.*

Proof. Every node of every n-node binary tree has depth at most $n-1$, and the counted visits for an insertion are at most its insertion depth. These deterministic upper bounds, together with Theorem 8, give the claimed matching bounds. □

The proof uses an infinite dense key order. The Python implementation compares keys at runtime, although its annotations say `int`; the rational model must therefore be distinguished from the annotated interface. For any fixed finite horizon, its finitely many possible adaptive histories use finitely many rational keys; an order-preserving rescaling embeds those keys in the integers. This yields the same finite-horizon worst-case bounds over integer inputs, with the spacing allowed to depend on the horizon. Comparisons here have unit cost; the integer spacing may require more than $O(\log n)$ bits per key, so the embedding does not establish the result for a fixed-width integer universe or account for arbitrary-precision arithmetic costs. The counterexample does not prove a linear lower bound for a fixed, nonadaptive insertion sequence.

5 An entropy-optimal alternative sampler

The following alternative changes only bit consumption, not the target law: read bits until the first zero or until k bits have been read, then return $T = k - m$ where m is the number of leading ones read. When $k = 0$, return zero without reading a bit. The original sampler requests one more bit whenever its first k bits are ones, although the target is already known.

Theorem 10 (Optimal expected bits for this target law)**.** *The alternative has exactly the target distribution in Theorem 3 and uses* $2 - 2^{1-k}$ *expected bits. It attains the entropy* $H(T)$*, hence no exact fair-bit prefix sampler for one independent target can use fewer bits in expectation without recycling randomness from previous or future samples.*

Proof. The alternative stops at the prefix $1^{k-t}0$ for $1 \le t \le k$ and at 1^k for $t = 0$. These prefixes have the probabilities stated in Theorem 3, and their lengths are $-\log_2 \mathbb{P}(T = t)$. The expected length is $\sum_{j=0}^{k-1} 2^{-j} = 2 - 2^{1-k}$, also valid as an empty sum for $k = 0$; it equals $H(T)$.

For the lower bound, let S be the random stopping prefix of any exact almost-surely terminating fair-bit sampler. Its possible prefixes are prefix-free, and a prefix s of length $|s|$ occurs with probability $2^{-|s|}$. Hence $H(S) = \mathbb{E}|S|$ (also in the extended sense). Since the output is a deterministic function of S, $H(T) \le H(S)$ by entropy's chain rule; see Shannon [3]. Thus the alternative meets the bound. This is optimal for the *given output distribution*, not a lower bound on all possible balancing algorithms. Bit recycling across outputs changes the model. □

The alternative is a mathematical variant, not the implemented sampler. Even with the same pseudorandom seed it generally changes subsequent draws and tree shapes. The repository's stored-priority treap reference counts 32 logical bits per successful insertion and stores a priority per key; it is not the same model as ideal continuous-priority treaps [2].

6 Depth lower bound and research status

Theorem 11 (Universal logarithmic lower bound)**.** *For any distribution on binary trees containing the same* $n \ge 4$ *distinct keys, there exists a key whose expected depth is at least* $\frac{1}{2}(\lfloor \log_2 n \rfloor - 1)$*.*

Thus a uniform bound on all nodes' expected depth cannot be $o(\log n)$, even with arbitrary balancing metadata and rotations.

Proof. Put $h = \lfloor \log_2 n \rfloor - 1$. Every binary tree has at most $2^h - 1 < n/2$ vertices at depths below h. Hence at least $n/2$ vertices have depth at least h, so the average of all n depths is at least $h/2$ in every realization. Take expectations and interchange the finite sum with expectation. At least one of the n fixed keys has expected depth at least the average. □

Property	Status for the implemented strategy
BST and key preservation	Proved, Theorem 1.
Node balance metadata	None; a global count and external random state are used.
Rotations	At most two per successful insertion, deterministically.
Random bits	Fewer than two in expectation, Theorem 3.
Locality	Uses n and may traverse far above the inserted node.
Expected repair time	Adaptive sequences can require $\Omega(n)$ expected visits by Theorem 8; the fixed-sequence bound is open.
Expected depth of every node	$O(\log n)$ for monotone sequences by Theorem 7, but $\Omega(n)$ is possible under adaptive keys by Theorem 8. The general fixed-sequence case is open.
Optimality	The logarithmic lower bound is achieved for monotone sequences but not against adaptive insertions.

The guarantees in the first four rows hold for fixed and adaptive histories under the stated fresh-bit assumption. The counterexample directly analyzes the target-depth rule; Brodal [1] exhibits related linear-depth behavior for different bottom-up random repair rules. Constant expected repair needs the excess-depth estimate in Theorem 4. Finite benchmarks and bounded enumeration can suggest conjectures or refute finite universal claims, but do not establish asymptotic bounds.

References

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